PS-3: How do we predict a location, and how sure can we be?

An explanation and brainstorming document for the team

> NOTE
>
> **What this document is for:** explaining ideas and comparing options. It does not contain code or a software design. Every number labelled *"from our data"* was computed directly from the PS-3 / shared CSVs. Nothing here uses invented columns.

0. The problem on one page, and what the data already tells us

The task: for each address, output **(a)** a predicted point (x, y)(x, y) and **(b)** a radius rr with an honest promise, for example: *"the real door is inside this 80% circle 80% of the time."*

The coordinates: `x`, `y` are on a local grid measured in metres. So the straight-line distance between two points is simply the error in metres.

Facts from our data that shape everything below

#	Fact (from our data)	Why it matters
F1	Only **100** addresses have true coordinates (`surveyed_addresses.csv`). Under `splits.csv` that is **66 train / 19 validation / 15 test**.	Every model must work with very little "answer key" data. Complex ML trained only on these 66 rows will overfit.
F2	Geocoder error depends heavily on `precision` . Median error: street **109 m**, locality **386 m**, pincode **1,376 m**. Rooftop has only **1** surveyed example.	Precision is the strongest clue for **how wide the prior should be**. We cannot learn anything reliable about "rooftop" from 1 example.
F3	Most geocodes are `locality` (2,052 of 2,880). Only 50 are `rooftop` .	For a typical address, the geocoder alone is off by about 400 m.
F4	**1,477 of 3,117** addresses have at least one visit. Visited addresses have about 3–4 visits each (median 3, maximum 9). Of the 100 surveyed addresses, **49** have visits and **51** have none.	About half the time we must predict with **no visits**. This is a core case, not a rare one.
F5	Visits where the agent reached the door are very close to the truth. Median distance to the true location: `met_family` **16 m**, `locked_premises` **20 m**, `neighbour_says_shifted` **20 m**, `met_borrower` **34 m**.	A good visit is about **20× more accurate** than a locality-level geocode.
F6	**But** the same outcomes have huge tails. The 90th percentile for `locked_premises` is **3,866 m**, and some `met_borrower` visits are **7.9 km** off.	Errors are **not bell-shaped**. Most visits are excellent and a few are wildly wrong. This is the main reason to question a pure Gaussian model.
F7	`address_not_traceable` check-ins are a median **1,603 m** from the truth.	This is where the agent *gave up*, not where the house is.
F8	Visit error grows with `gps_accuracy_m` . Visits in the best quarter of accuracy have a median error of 17 m, and the worst quarter 53 m. The median ratio of error to `gps_accuracy_m` is **≈ 2.4**.	GPS accuracy is a useful signal, but **error ≠ accuracy_m**. Treating them as equal would make us overconfident.
F9	172 visits share a `photo_hash` with another visit. Agent FA009 has the worst median door-visit error (62 m vs 15–47 m for the others).	Some visits are probably not genuine. We need a trust score.

WARNING

Honest status of our current prototype. Our code currently (i) sets $\sigma_i \approx$ σi ≈ `gps_accuracy_m` , which F8 suggests is too small by about 2.4×; (ii) uses a random 50/50 split of the 100 surveyed rows instead of `splits.csv` ; (iii) uses fixed prior widths per precision level that resemble error statistics from *all* 100 surveyed rows, so the test set may have leaked in. Under-coverage (60% for the 80% target) is what an overconfident σσ would produce. The 70.6 m median error should be treated as provisional until we re-run it on the official split.

2. What our current Bayesian approach is trying to do

2.1 In plain words first

Picture the town as a grid of small squares, each 25 m × 25 m.

1. **Before looking at visits**, we give every square a score for "how likely is the house here?" using what we already know: the geocoder pin and how precise it claims to be. This is the **prior**.
2. **For each visit**, we ask of every square: *"If the house really were in this square, how natural would this visit's GPS reading look?"* Squares close to a trustworthy check-in get a high score. Far squares get a low score. This is the **likelihood**.
3. **We multiply** the prior by all the visit scores, square by square, then rescale so all squares add up to 1. The result is the **posterior**: our updated belief after seeing all the evidence.
4. The best guess is the **peak** of the posterior. The confidence circle is the **smallest region around that peak that holds 80% (or 90%) of the belief**.

2.2 Mathematically

$$\underbrace{P(L \mid E)}_{\text{posterior}} \propto \underbrace{P(L)}_{\text{prior}} \prod_{i=1}^{N} \underbrace{P(V_i \mid L)}_{\text{likelihood of visit } i}$$

```
posterior
P(L | E) ∝ prior
P(L) i=1∏N likelihood of visit i
P(Vi | L)
```

Symbol	Meaning	In our problem
LL	A **candidate location**: one possible place the house could be	One grid square (x, y)(x, y)
EE	**All evidence** for this address	Geocoder pin + precision + all visits (+ upstream address features, landmarks, localities)
V_iVi	One visit's observation	Check-in point `checkin_x`, `checkin_y` and its attributes
$P(L)$P (L)	**Prior**: belief before visits	Built from the geocoder (and, later, other non-visit clues)
$P(V_i \mid L)$P (Vi ∣ L)	**Likelihood**: how plausible visit ii is *if* the house were at LL	Depends on distance from LL to the check-in, GPS accuracy, and trust
$P(L \mid E)$ P (L ∣ E)	**Posterior**: belief after all evidence	The final heat map
$\propto$ ∝	"proportional to"	We rescale at the end so the probabilities add up to 1

What the formula means in our problem: *"A square is a good guess if the geocoder thought it plausible AND every visit is consistent with it."*

2.3 Why are the likelihoods multiplied?

The word "AND" becomes multiplication. If visit 1 says "probably near here" (0.8) and visit 2 independently says "probably near here" (0.7), the joint support is $0.8 \times 0.70.8 \times 0.7$.

Multiplication has one important property: **a single visit that strongly says "not here" (a value near 0) can veto a square.** This is why we need a trust or outlier term (§3.3). Without it, one fake visit could wipe out the true location.

2.4 Assumptions hidden in the formula

Assumption	Plain meaning	Is it realistic for us?
Conditional independence	Given the true house location, one visit's error does not affect another's	⚠️ Partly false. The same agent visiting the same address repeatedly makes *correlated* errors. FA009 can produce several near-identical fake check-ins, and multiplying them counts the same lie many times.
One true location	The address has one door	Mostly yes. But "office" (464) and "permanent_native" (226) address types may behave differently.
The likelihood shape is correct	We know how check-ins scatter around a house	This is the big question (§4–5)
The prior is honest	Geocoder uncertainty is described correctly	Only if it is calibrated per precision level on training data

2.5 Mapping every term to our files

Component	Source CSV	Actual columns	How certain is it?
Prior centre	`baseline_geocodes (1).csv`	`geocoder_x`, `geocoder_y`	Off by 100 m to 1.4 km (F2)
Prior width	`baseline_geocodes (1).csv`	`precision`	Category only. The width has to be *learned*
Town extent	`towns.csv`	`approx_radius_m`	A coarse limit on where the house can be
Locality centre	`localities.csv`	`centroid_x`, `centroid_y`	Usable **only once upstream tells us which locality** an address belongs to. `addresses.csv` has no locality column
Landmarks	`landmarks_poi.csv`	`x`, `y`, `landmark_type`	Exact points. Linking a landmark to an address needs upstream features
Visit location	`field_visits.csv`	`checkin_x`, `checkin_y`	Very good to useless, depending on outcome and trust
Visit precision	`field_visits.csv`	`gps_accuracy_m`	Self-reported by the phone. Correlated with error, but not equal to it
Visit meaning	`field_visits.csv`	`outcome`, `remark`, `dwell_s`	Recorded by the agent, and may be false
Trail	`visit_gps_points.csv`	`x`, `y`, `seq`, `accuracy_m`, `point_ts`	Lets us derive path length, displacement and tortuosity
Agent	`agents.csv`	`agent_id`, `tenure_months`	Context for trust
Answer key	`surveyed_addresses.csv`	`surveyed_x`, `surveyed_y`	Treated as truth. Only 100 rows

3. Where do the probabilities actually come from?

> IMPORTANT
>
> The reference PDF gives the **framework** (prior × likelihood, outlier mixture, conformal calibration). It does **not** give one fixed formula for building the prior, a fixed number for σ_i σi, or a specific trust model. **Those are design choices we must make and justify with data.** This is a strength for the report: we can show *how* we chose them.

3.1 The prior $P(L)$P (L)

Plain meaning: where we think the house is *before any agent goes there*.

A simple, defensible construction: a circular bump centred on the geocoder pin, with a width that depends on `precision`:

$$P(L) \propto \exp\left(-\frac{\|L-g\|^2}{2\sigma_{\text{prec}}^2}\right)$$

```
P(L) ∝ exp(−2 σprec2 ‖L − g‖2)
```

- gg is the geocoder pin, and σ_{prec} σprec is a width learned for each precision level.
- **What it means for us:** a `street`-level pin gives a bump about 100 m wide. A `pincode` pin gives a bump more than a kilometre wide.

How to learn σ_{prec} σprec honestly: on the **training** rows only, measure the geocoder errors for each precision level and choose σσ so the model's error spread matches what we see. Small detail: for a 2-D circular Gaussian, the *median radial* error is 1.177σ1.177 σ. So a median error of 386 m implies $\sigma \approx 328$σ ≈ 328 m per axis, not 386 m.

Other clues that could reshape the prior (beyond a simple circle):

Clue	How it could enter the prior	Data status
Geocoder + precision	Centre + width	✅ Available
Town	Remove (or heavily penalise) squares outside the town radius	✅ Available
Locality (from upstream)	A second bump at the locality centroid. The prior becomes "near the geocoder AND inside the stated locality"	⏳ Needs upstream mapping
Landmark (from upstream)	"Behind the temple" → raise squares within about 150 m of the matching landmark point	⏳ Needs upstream mapping
Nearby confirmed addresses	Neighbouring addresses with trusted door visits show where houses *actually* are around this geocode. The prior can use their pattern (for example, whether the geocoder is consistently shifted in that area)	🔧 Derivable from visits of *other* addresses
Address type	`office` vs `residence` vs `permanent_native` could need different widths	✅ Available in `addresses.csv`

> NOTE
>
> The 237 addresses missing from the geocoder file (mostly `permanent_native`) have **no geocoder prior**. For these, the prior can only come from the town, the locality, or a very wide default.

3.2 The likelihood $P(V_i \mid L)$P (Vi ∣ L)

Plain meaning: *"If the house were at LL, how believable is it that the agent's phone reported this check-in point?"*

Numerical example. Suppose a genuine visit has scatter $\sigma = 20$σ = 20 m. Using the Gaussian score $\exp(-d^2/2\sigma^2)$ exp(−d2/2σ2):

Candidate square at distance dd from check-in	Score
$d = 10$d = 10 m	$e^{-0.125} \approx 0.88$e−0.125 ≈ 0.88
$d = 40$d = 40 m	$e^{-2} \approx 0.14$e−2 ≈ 0.14
$d = 100$d = 100 m	$e^{-12.5} \approx 0.000004$e−12.5 ≈ 0.000004

So this visit says: *"very likely within about 40 m of me, and almost certainly not 100 m away."*

Worked fusion example (prior + one visit). The geocoder is `locality` ($\sigma_{\text{prior}} \approx 330$σprior ≈ 330 m) and there is one good visit with $\sigma_{\text{visit}} = 40$σvisit = 40 m. When two Gaussian sources are multiplied, the result is weighted by $1/\sigma^2$ 1/σ2:

$$w_{\text{visit}} = \frac{1/40^2}{1/40^2 + 1/330^2} \approx 0.986, \qquad \sigma_{\text{post}} = \left(\frac{1}{40^2} + \frac{1}{330^2}\right)^{-1/2} \approx 39.7 \text{ m}$$

```
wvisit = 1/402 + 1/33021/402 ≈ 0.986,    σpost = (4021 + 33021)−1/2 ≈ 39.7 m
```

What this means for us: one good visit pulls the answer about 99% of the way from the geocoder to the visit, and shrinks the uncertainty from about 330 m to about 40 m. This matches F2 and F5: visits are far more precise than locality geocodes.

3.3 Visit trust π_i πi

$$\pi_i = P(\text{visit } i \text{ is genuine})$$

```
πi = P(visit i is genuine)
```

The PDF's mixture likelihood uses it like this:

$$p(z_i \mid \mu) = \pi_i N(z_i; \mu, \sigma_i^2 I) + (1 - \pi_i)\text{Outlier}(z_i)$$

```
p(zi | μ) = πiN(zi; μ, σi2I) + (1 − πi) Outlier(zi)
```

Plain meaning: *"With probability π_i πi this check-in is a real measurement near the house. Otherwise it is junk, which could have landed anywhere."* The outlier term is a small, flat value. It stops a junk visit from vetoing the true location, because the score never drops to exactly zero.

Critical distinction:

	Recorded visit outcome	Model-derived trust π_i πi
What it is	A **column in the data** (`outcome`)	A **number we compute**
Who made it	The field agent	Our model
Example	`met_borrower`	0.93
Can it be wrong?	Yes. An agent can log `met_borrower` from a tea stall	It is an estimate, with its own error
Role	An **input** to trust	An **output** used in the likelihood

Signals that can influence π_i πi (all derivable from our files):

Signal	Source	Suspicious when…
`outcome`	field_visits	`address_not_traceable` (the check-in is not at the house, F7)
Photo hash reuse	field_visits `photo_hash`	The same photo appears for different addresses
Agent hotspot	field_visits `agent_id` + check-in	One agent logs many *different* addresses from the same 25 m cell
Trail displacement and tortuosity	visit_gps_points	The phone barely moved, or the path is physically odd
`dwell_s`	field_visits	The dwell time is implausibly short for the claimed outcome
Remark vs outcome contradiction	field_visits `remark`	The outcome says "met" but the remark says "locked"
Agreement with other visits	field_visits	The visit is far from where other agents check in for the same address
`tenure_months`	agents	Weak context only

> CAUTION
>
> **We have no fraud labels.** Any trust model we build is a **reliability score**, not a proven fraud detector. A cleaner way to tune it: on the training surveyed addresses, call a visit "consistent" if it lands within, say, 100 m of the true location. Then learn which signals predict consistency. That is a real, measurable target, but with few examples, so keep the model simple.

3.4 Visit uncertainty σ_i σi

What uncertainty means: if the agent stood at the same door 100 times, the check-ins would scatter. σ_i σi describes **how wide that scatter is**.

Why visits differ:

- The phone's GPS fix quality varies (open sky vs a narrow lane).
- The agent may stand at the lane entrance rather than at the door.
- The outcome affects position. "Met family" means the agent was at the door. "Neighbour says shifted" means a neighbour's door, which is close but not identical.

Why `accuracy_m` ≠ σ_i σi:

1. `accuracy_m` is the phone's *own* guess of GPS error. It does not include "the agent stood 20 m away from the house."
2. Our data shows the median error is about **2.4×** `gps_accuracy_m` (F8).
3. Phones define accuracy differently (often a 68% radius, not a standard deviation).

How to estimate it from data (conceptual): on training addresses with door outcomes, compute each visit's true error e_i ei. Then fit a simple relationship such as

$$\sigma_i = \sqrt{a^2 + (b \cdot \text{accuracy}_i)^2}$$

```
σi =  a2 + (b · accuracyi)2
```

- aa is a floor: "even perfect GPS is about aa metres from the door."
- bb is a scale on reported accuracy.
- Check that the fitted σσ reproduces the observed error quantiles in each accuracy bin (F8's table gives a first look).
- **What this means for us:** probably $b > 1$b > 1 and aa around 10–15 m.

4. Gaussian, Bayesian, sigma and radius: four different jobs

These ideas are often blurred together. They do separate jobs:

Concept	Its single job	Analogy
Bayesian inference	A **rule for combining** evidence into one belief	The *recipe*
Gaussian distribution	One particular **shape** for how errors scatter (a bell / round bump)	One *ingredient* choice
Sigma (σσ)	The **width** of that shape. Bigger means more spread out	How much of the ingredient
Confidence radius (r_α rα)	A **summary of the final belief**: the circle holding αα of it	The *serving size* we promise

Key point: Bayes' rule does **not** require Gaussians. Gaussians are popular because the maths becomes neat: precisions add, and you get the closed-form radius in §9. Any shape can be used on a grid.

5. Is a Gaussian a good assumption for THIS dataset?

5.1 The evidence says: not on its own

Data characteristic	What a Gaussian assumes	What we actually see
Visit errors	Almost never beyond 3σ	`met_borrower` median 34 m but maximum 7.9 km. That is hundreds of σ away
Locked premises	Same	Median 20 m, 90th percentile 3.9 km. Most are great, and some are completely elsewhere
Suspicious visits	Not modelled	172 reused photos, and an agent hotspot pattern
Conflicting visits	One bump	Two groups of check-ins = **two bumps** (multimodal)
Geocoder by precision	One σ	Different σ for each level, and pincode errors are very skewed (median 1.4 km, maximum 4.8 km)
No-visit addresses	—	Belief comes only from the prior, which is very broad

Conclusion: a Gaussian is a reasonable description of a **genuine** visit's scatter (the core), but **not** of the full error distribution, which has a **heavy tail**. The PDF's mixture term $(1-\pi_i)\text{Outlier}$(1 – πi) Outlier exists precisely to patch this.

5.2 Alternatives (conceptual only)

Alternative	Why it could be better	Problem it solves	What it assumes	Realistic for us?
Student-t likelihood	Bell-shaped core with **fat tails**. Far points lose influence smoothly	Outliers without needing a trust score	One extra "tail heaviness" parameter	✅ Very good fit for F6. Simple
Gaussian + outlier mixture (PDF)	Explicitly says "real or junk". Uses π_i πi	Fake visits, lost agents	Junk is roughly uniform over the town	✅ Our current choice. Good *if* π_i πi is decent
Empirical distribution	Uses the *actual* histogram of training errors (for each outcome) as the likelihood shape	No shape assumption at all	Enough training errors (we have about 150 door visits on surveyed addresses)	⚠️ Feasible for broad groups only
KDE (kernel density)	Smooth "heat" around every check-in, then add them up	Naturally multimodal	Kernel width choice	✅ Useful for visualising and finding clusters
Grid posterior	Computes the posterior on squares, so any shape works	Multimodality, odd priors (town edges, localities)	Grid resolution (25 m is fine)	✅ Already in use. It is what makes the alternatives possible
Particle posterior	A cloud of sample points instead of a grid	Same as grid, scales better	Enough particles	➖ Not needed at town scale
Robust likelihood (Huber, etc.)	Quadratic penalty near the centre, linear far away	Limits the damage of outliers	A threshold choice	✅ Similar in spirit to Student-t

What this means for us: keep the **grid**, which handles any shape, and replace the pure Gaussian visit term with **heavy-tailed or mixture** terms whose parameters are fitted on training errors.

6. Better or alternative ways to predict the LOCATION itself

A. Robust spatial estimators (visits only)

Estimator	Idea	Helps when	Fails when
Mean	Average of the check-ins	All visits are good	One bad visit drags it away
Coordinate median	Median of x and median of y separately	Up to about half the visits are bad	Can produce a point that is not "between" the visits
Geometric median	The point minimising the *sum of distances*	Same, but rotation-invariant and more natural in 2-D	More than half the visits are bad
Weighted geometric median	Same, with weights π_i/σ_i πi/σi	Trust and accuracy vary	Weights are badly estimated
RANSAC	Repeatedly pick a small subset, count how many visits agree within a radius, keep the best consensus	A clean cluster plus random junk	Too few visits (median is only 3)
M-estimators	Down-weight points with large residuals, then iterate	Moderate outliers	Many outliers

Example: 4 visits near the house at about (1000, 1000), and 1 fake visit at (3000, 1000). The mean x = 1400 is **400 m off**. The median x ≈ 1000, which is correct.

B. Clustering-based approaches

DBSCAN: "points with at least `k` neighbours within `ε` metres form a cluster. Isolated points are noise."
HDBSCAN: the same idea, but it tries many values of `ε` and keeps the most stable clusters. You do not have to choose `ε` yourself.

```
    Visit5 ×                                  (2 km away → noise)

                    ×  ×
                  ×  🏠        ← Visits 1–4 form a cluster
```

1. The cluster with 4 members wins, and the lone visit is labelled noise.
2. The prediction is the centre of the winning cluster.
3. With **two** clusters (for example, the old house vs the new house after "shifted"), clustering *reveals* the conflict instead of averaging into an empty field between them.

Caveat for us: most visited addresses have only 1–3 visits, which is often too few to form clusters. Clustering helps most at the 5–9 visit end.

C. Residual learning (correcting the geocoder)

```
Address → Commercial geocoder → Baseline pin g → Predict correction (Δx, Δy) → Final = g + Δ
```

Why predicting the correction is easier: the geocoder already gets you to within about 400 m. Learning "the geocoder is usually 200 m too far north in this part of T2" is a much smaller, more structured problem than learning absolute coordinates from scratch.

Features that could plausibly predict the correction:

- `precision`, `town_id`, `address_type`
- Geocoder pin minus the locality centroid (from upstream locality + `localities.csv`)
- Distance and direction to the nearest landmark of the type mentioned (from upstream)
- **Local geocoder bias**: the average (trusted visit – geocoder) offset for *neighbouring* addresses

> TIP
>
> **Biggest opportunity: pseudo-labels.** We have only 66 true training points, but about 1,477 visited addresses. For addresses with **several agreeing, trusted door visits**, the visit consensus is accurate to about 20–30 m (F5). That is good enough to act as an approximate "truth" for training a geocoder-correction model, which could grow the training set roughly 10×. The 66 surveyed training rows are then used to *check* the pseudo-labels, not to train. Risk: pseudo-labels inherit any bias in the visits, such as FA009's pattern.

D. Gradient boosting (XGBoost / LightGBM / CatBoost)

- **Fits well:** tabular, mixed numeric and categorical features (CatBoost handles categories like `precision` and `town_id` natively), and modest data sizes.
- **Use it for:** (1) predicting $\Delta x, \Delta y$Δx, Δy (residual learning), (2) the visit trust score π_i πi, (3) predicting the *expected error size* per address. That third use gives a raw radius that conformal calibration can then adjust.
- **Do not:** train it only on 66 rows with many features. Use pseudo-labels, or keep it very shallow and regularised.
- **Output:** boosting gives a *point*. It does not give a probability map on its own. For a radius it needs quantile regression or conformal calibration around it.

E. Hybrid approaches

```
[Geocoder pin] --(ML correction)--> [Better prior centre + learned width]
                                                  ×
[Visits] --(trust + robust/heavy-tailed likelihood)--> [Visit evidence]
                                                  ×
[Town / locality / landmark limits] -------> [Spatial context]
                                                  ↓
                                    Grid posterior → pin + raw radius → conformal
```

The idea: each component fixes a different weakness.

- **ML correction** improves the 51% of cases with no visits.
- **Robust visit terms** fix the tail problem.
- **Spatial context** removes impossible areas.
- **Conformal calibration** makes the radius honest whatever errors remain.

The Bayesian grid is simply the **glue** that combines them consistently.

7. Messy and conflicting visits: how a sensible method should behave

Case	What the evidence means	Sensible treatment
1. One good visit, close	Strong evidence near the true house (F5: about 20 m typical)	Posterior moves almost fully to the visit. Radius shrinks from hundreds of metres to tens. Still keep a tail: one visit could be the 10% bad case
2. Several good visits cluster together	Independent confirmation (ideally from different agents)	Very narrow posterior. Be careful: 4 visits by the *same* agent are not 4 independent confirmations. Consider limiting each agent's total weight
3. One visit far away	Either the house is wrong or the visit is junk	The heavy-tailed or mixture likelihood lets the far point "explain itself" as an outlier, so the cluster wins. If it is the *only* visit, weigh it against the prior. A good visit with high trust still usually wins
4. Unsuccessful visit (`address_not_traceable`)	The agent could **not** find the house. The check-in is where they gave up (median 1.6 km away, F7)	Treat it as **(almost) uninformative about location**, or at most a weak "the agent searched around here" signal. Do **not** treat it as a house sighting. It is a strong signal that the *address text* is ambiguous, so the radius should stay **wide**
4b. `locked_premises`, `neighbour_says_shifted`	Failed for collection, but the agent **was at the door** (median 20 m)	Treat it as **good location evidence**. This is a key insight for the report: collection failure ≠ location failure
5. Poor GPS accuracy	The phone itself is unsure	Larger σ_i σi (via the learned mapping). The visit still counts, but it pulls less and narrows the radius less
6. Different agents disagree	Possible multimodality: a wrong house, a shifted borrower, or one agent faking	Use the grid to keep **both bumps**. Let trust decide. If it is still ambiguous, the radius must cover both modes or be reported as "two candidates." Never silently average them into the empty space between
7. Suspiciously similar locations	Many *different* addresses checked in from the same spot (an agent hotspot), or reused photos	Trust drops sharply. These visits should barely move the posterior. Repeated identical points must not count as independent confirmations
8. No visits	Only non-visit evidence	Use the prior alone (§8). The radius should be wide, and its width should depend on prior quality

8. The no-visit case (about half of all addresses)

Available evidence: geocoder pin + `precision`, town, address type, and (once upstream is ready) locality and landmark features. Neighbourhood patterns from *other* addresses' trusted visits can also help.

"No visits" does not mean "no confidence interval." It means *a wider one*. From our data, a `street` geocode alone already supports roughly a 140–170 m radius at 80–90%. A `locality` geocode supports about 540–630 m, and a `pincode` geocode needs several kilometres. These are honest statements, and the business can use them.

How uncertainty should behave:

Evidence strength	Example	Expected radius
Strongest	Several trusted, agreeing door visits	Tens of metres
Strong	One trusted door visit	About 50–100 m
Medium	Street geocode, no visits	About 150 m
Weak	Locality geocode, no visits	About 500–650 m
Weakest	Pincode geocode, or no geocode at all	Kilometres / most of the town

Ways to improve no-visit predictions: a residual correction (§6C), locality and landmark constraints from upstream, and local geocoder bias learned from neighbours' visits.

9. The confidence radius in depth

9.1 The definition

$$P(\|\text{true} - \text{pin}\| \le r_\alpha) = \alpha$$

P (‖true − pin‖ ≤ rα) = α

Plain meaning: *"Draw a circle of radius r_α rα around our pin. Across many addresses, the real door lands inside it a fraction αα of the time."* For $\alpha = 0.8$α = 0.8: 8 out of 10 addresses.

Concept	Question it answers
Location prediction	Where is the house? → a point
Uncertainty	How unsure are we? → a spread (σ, or the posterior's shape)
Confidence radius	How big an area must we hand over to keep our promise? → one number in metres

9.2 The Gaussian formula

If the error in x and the error in y are each Gaussian with the **same** σ, **independent**, and **centred on zero**, then the radial error follows a **Rayleigh** distribution, and

$$r_\alpha = \sigma\sqrt{-2\ln(1-\alpha)}$$

```
rα = σ  −2 ln(1 − α)
```

αα	Multiplier	Example: σ = 40 m
0.50	1.18 σ	47 m
0.80	1.79 σ	72 m
0.90	2.15 σ	86 m
0.95	2.45 σ	98 m

Assumptions behind the formula (all can fail for us):

1. **Circular**: same spread in x and y. This fails if, for example, the error runs along a road.
2. **Unbiased**: the pin is centred on the truth on average. This fails if the geocoder is systematically shifted in some area.
3. **Gaussian tails**: this fails because of F6.
4. **σ is known exactly**: in reality we estimated it, sometimes from a single number like `precision`.
5. **One mode**: this fails with conflicting visits.

So the formula is a **useful first guess**, not a guarantee. On a grid posterior we do not even need it: we read the radius directly as "the circle holding 80% of the posterior mass." Both versions then still need §10.

10. Conformal calibration

10.1 Why we need it even with a Gaussian radius

The model says: *"80% radius = 400 m."* That claim is only as true as the assumptions behind it. If σ was too small (F8 suggests it is), or tails are heavy (F6), the true house might land inside only 60% of the time. **That is exactly what our prototype showed.**

Conformal calibration **checks the promise against reality and fixes it**, using held-out true locations.

10.2 The idea in 4 steps (split conformal)

1. Keep a **calibration set**: surveyed addresses *not* used for fitting (for us, the 19 `validation` rows).
2. For each one, compute a **score** = actual error ÷ raw model radius. A score of 1.5 means "the truth was 1.5× further away than our radius."

3. Find the value qq such that about 80% of scores are below it. More precisely, the $\lceil (n+1)\alpha \rceil$⌈(n + 1)α⌉-th smallest score.
4. For every new address: **final radius =** q**q × raw radius**.

Example: if 80% of calibration scores are ≤ 1.6, every raw radius is stretched by 1.6×. If the model was too *cautious*, $q < 1$q < 1 shrinks the circles.

The guarantee: if new addresses are "like" the calibration ones (exchangeable), coverage will be at least αα on average. This holds regardless of whether the Gaussian, the trust model or any other part was correct.

Term	Meaning
Calibration set	Held-out addresses with truth, used only to measure qq
Held-out errors	Real distances (pin to truth) on those addresses
Coverage	% of test addresses whose truth falls inside the circle. Should be about αα
Sharpness	How *small* the circles are. A 5 km circle always "covers" but is useless. We want coverage met with the **smallest** radius
Split conformal	One shared multiplier qq for everyone
Mondrian / group-conditional	A separate qq for each group (for example, "has visits" vs "no visits", or per precision level), so coverage holds *within* each group, not just on average

WARNING

Small-data reality: with 19 calibration rows, qq is noisy. With 15 test rows, one address = 6.7 percentage points of coverage. Mondrian groups get even smaller (rooftop has 1 surveyed row). Options: group coarsely (visits vs no visits); combine train + validation with cross-validation ("CV+ / jackknife+" conformal); and report coverage **with confidence intervals** (for example, 12/15 = 80% has a plausible range of about 55–93%).

11. Comparing the candidate approaches

Method	What it predicts	Main assumption	Handles outliers?	Handles multiple clusters?	Works with small data?	Interpretability	Suitability for PS-3
Commercial geocoder	Point	The geocoder is right	—	—	✅ No training	✅ High	Baseline only (median about 380 m)
Visit mean	Point	All visits are good	❌	❌	✅	✅	Weak
Median / geometric median	Point	Fewer than 50% of visits are bad	✅	❌	✅	✅	Strong simple baseline, but visits only
Weighted geometric median	Point	Trust and σ are correct	✅✅	❌	✅	✅	Strong for visited addresses
DBSCAN / HDBSCAN	Point(s) + noise labels	Enough visits to form clusters	✅	✅	⚠️ Needs ≥3–4 visits	✅	Useful for the 5–9 visit cases and for diagnosis
Bayesian + Gaussian	Posterior → point + radius	Gaussian errors, independence	❌	⚠️ Only on a grid	✅	✅	Clean framework, but overconfident on our data
Bayesian + robust / mixture likelihood	Posterior → point + radius	Correct tail / trust model	✅✅	✅ (grid)	✅ Few parameters	✅ Medium-high	**Best core candidate**
Residual ML (boosting)	Δ correction → point	Geocoder errors are predictable from features	–	❌	⚠️ Only with pseudo-labels	Medium (feature importances)	Main hope for **no-visit** cases
Hybrid (ML prior + robust Bayes + conformal)	Point + calibrated radius	The parts combine sensibly	✅✅	✅	✅ If each part stays simple	Medium	**Most complete answer** to PS-3
Neighbour-based prior (spatial smoothing / kriging-like)	Prior centre / bias	Geocoder errors are spatially correlated	–	❌	✅ Uses other addresses' visits	✅	Promising add-on for no-visit cases
Quantile regression on error size	Radius directly	Error size is predictable from features	–	❌	⚠️	✅	Good input to conformal (adaptive radius)

Main lesson: no single method wins everywhere. **Robust visit handling** matters most for visited addresses. **Prior improvement** matters most for the 51% with no visits. **Conformal calibration** matters for *everyone's* radius.

12. What we should investigate next (recommendation)

> **Ground rule for all experiments:** fit on `train`, tune and calibrate on `validation`, report once on `test` (from `splits.csv`). Report median error, P90 error, coverage at 80/90% (with intervals) and median radius. Report them **separately for "has visits" vs "no visits"**.

① Calibrate the inputs: a learned σ_i σi mapping + a heavy-tailed or mixture visit likelihood

- **Why promising:** F8 (error ≈ 2.4× accuracy) and F6 (huge tails) directly explain our under-coverage.
- **Weakness fixed:** an overconfident σ, and pure-Gaussian tails.
- **Data:** train-split surveyed addresses with visits (about 30 addresses, about 100 door visits): `gps_accuracy_m`, `outcome`, check-ins.
- **Assumption:** the way errors depend on accuracy and outcome is stable across splits.
- **Experiment:** Gaussian with σ = σ = accuracy vs learned σσ vs Student-t vs Gaussian + outlier mixture. Compare **raw (pre-conformal) coverage** and the **conformal multiplier qq**. A qq closer to 1 means the model is honest by itself, which gives sharper final radii.

② A robust visit baseline as a reference point (weighted geometric median / HDBSCAN)

- **Why promising:** simple, explainable, and hard to beat for visited addresses. It tells us whether the Bayesian machinery actually adds value.
- **Weakness fixed:** we currently have no honest non-Bayesian comparison on the official split.
- **Data:** visits + trust signals only.
- **Assumption:** most visits for an address are genuine.
- **Experiment:** on visited test addresses, compare median and P90 error against Bayesian fusion. If they are equal, the Bayesian value lies mainly in the *radius* and the *no-visit fallback*, which is still a valid story for the report.

③ A better prior for no-visit addresses: precision-calibrated width + geocoder residual correction using visit pseudo-labels

- **Why promising:** half the addresses have no visits, and locality geocodes are off by about 400 m. Even a 20% improvement there moves the overall metrics noticeably.
- **Weakness fixed:** the prior is currently just "circle around the geocoder with a hand-set width."
- **Data:** `baseline_geocodes`, `addresses` (`town_id`, `address_type`), neighbours' trusted visit consensus as pseudo-truth, and later upstream locality/landmark features + `localities.csv` / `landmarks_poi.csv`.
- **Assumption:** geocoder errors have a learnable or spatially smooth pattern, and pseudo-labels are accurate to about 30 m.

- **Experiment:** on **no-visit** test addresses, compare raw geocoder vs local-bias-corrected geocoder vs boosted Δ correction. Check pseudo-label quality first on train surveyed addresses that also have visits.

④ Group-aware conformal calibration (visits vs no visits)

- **Why promising:** a single multiplier can over-cover visited addresses while under-covering no-visit ones, or the reverse.
- **Weakness fixed:** "80% on average" hiding "95% for one group, 60% for the other."
- **Data:** validation (or cross-validated train + validation) surveyed rows labelled by evidence regime.
- **Assumption:** each group has enough calibration rows. Use coarse groups only.
- **Experiment:** split vs Mondrian vs CV+ conformal. Report coverage **per group** and median radius per group. Prefer the method that meets coverage in both groups with the smallest radii.

The three questions, answered in one breath each

How do we predict the location? Start from the geocoder (widened according to its precision), let trusted door visits pull the belief toward where agents actually stood (weighted by GPS quality, robust to junk), and take the peak of the combined belief. When there are no visits, improve the starting point using learned geocoder corrections and spatial context.

How do we quantify and calibrate uncertainty? The spread of that combined belief gives a raw radius. Then we check it against held-out surveyed addresses and stretch or shrink it (conformal calibration) until 80% of true doors really fall inside the 80% circle, ideally checked separately for visited and unvisited addresses.

Why Bayesian + Gaussian, and what might be better? Bayes is the right *recipe* for combining a weak prior with several unequal visits. A Gaussian is a convenient *ingredient*, but our data has heavy tails and fake visits. So the Gaussian core should be wrapped in a mixture or Student-t likelihood with a σ learned from data, compared against simple robust baselines, and always finished with conformal calibration.